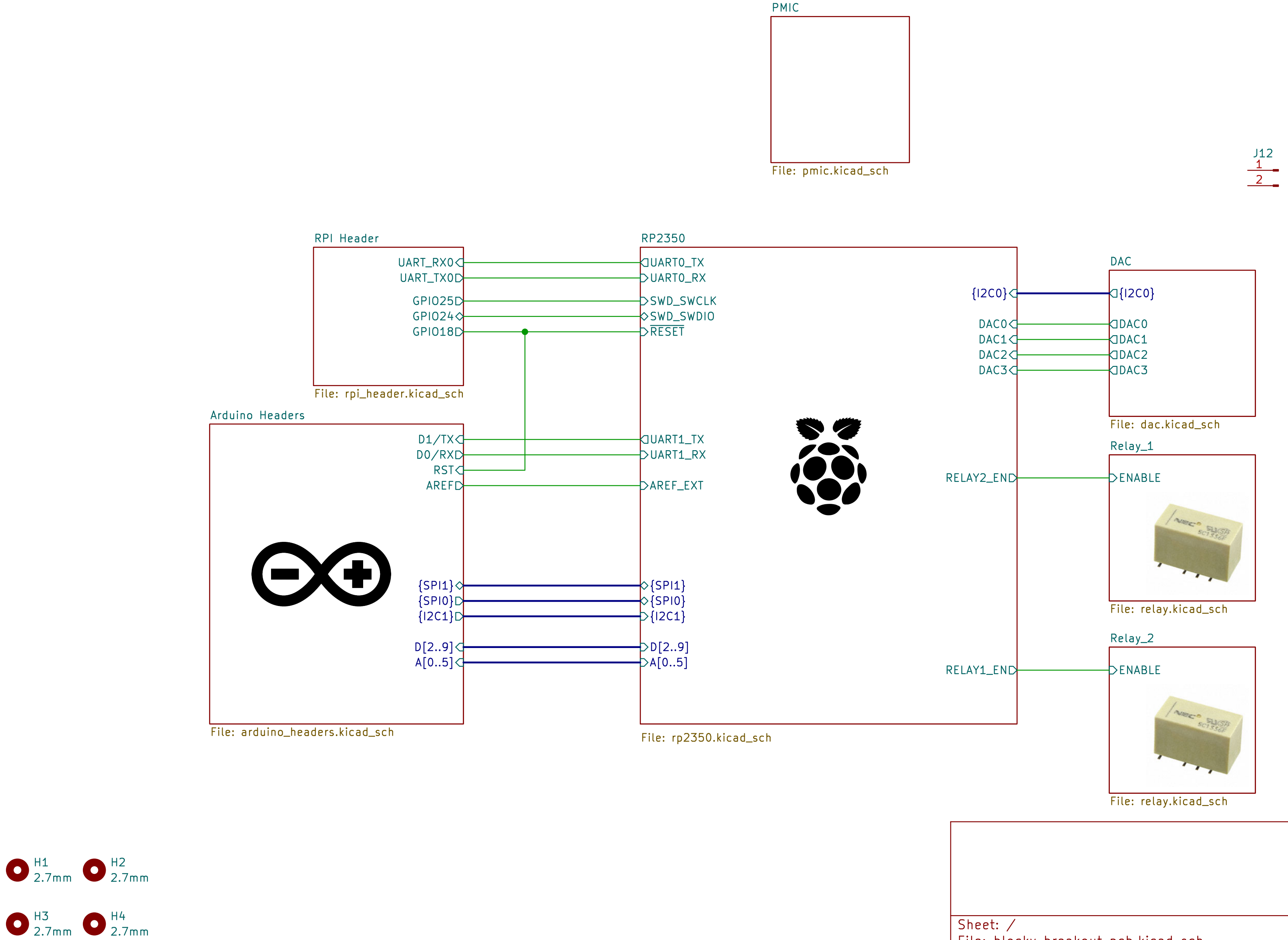
Ryan TODO list.
- Go over all pins numbers/names, make sure they're correct.
- Double check 3.3V supply component values
- Check all decoupling caps on the RP2350



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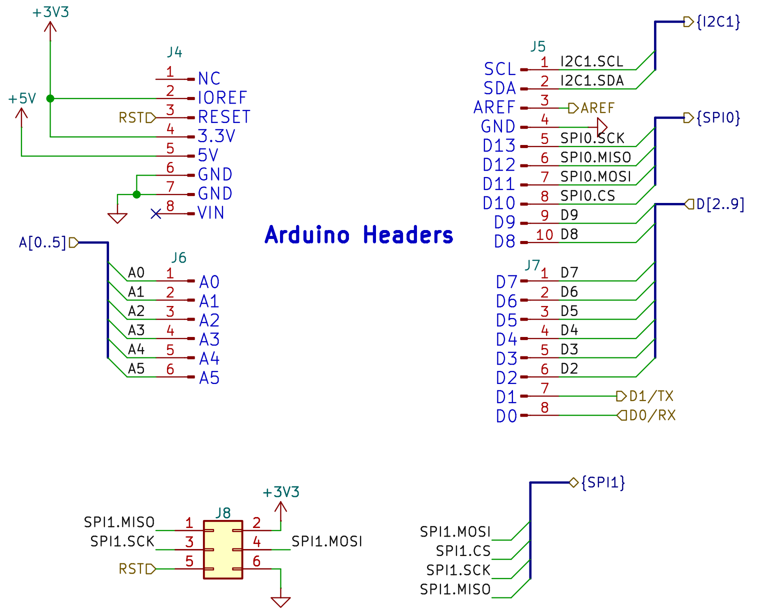
File: blocky_breakout_pcb.kicad_sch

Title: RPI <-> RP2350 Breakout

Size: A4 | Date: | **Rev: R0.1**
KiCad E.D.A. 9.0.3 | Id: 1/8



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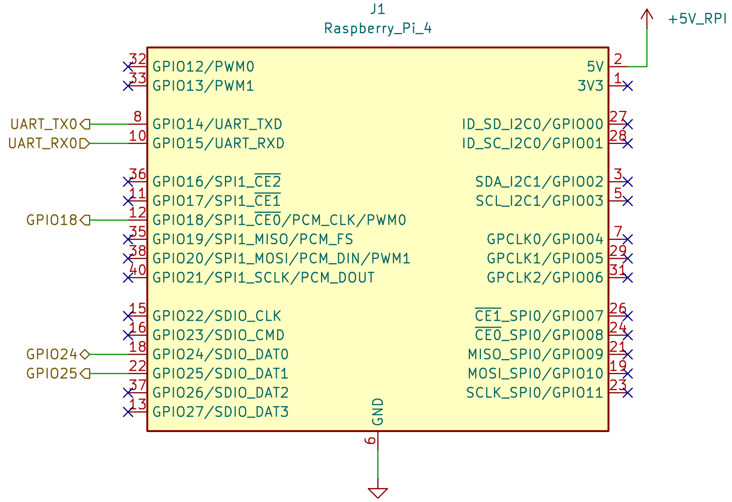
File: arduino_headers.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.3		Id: 5/8

Raspberry PI
Pin Header



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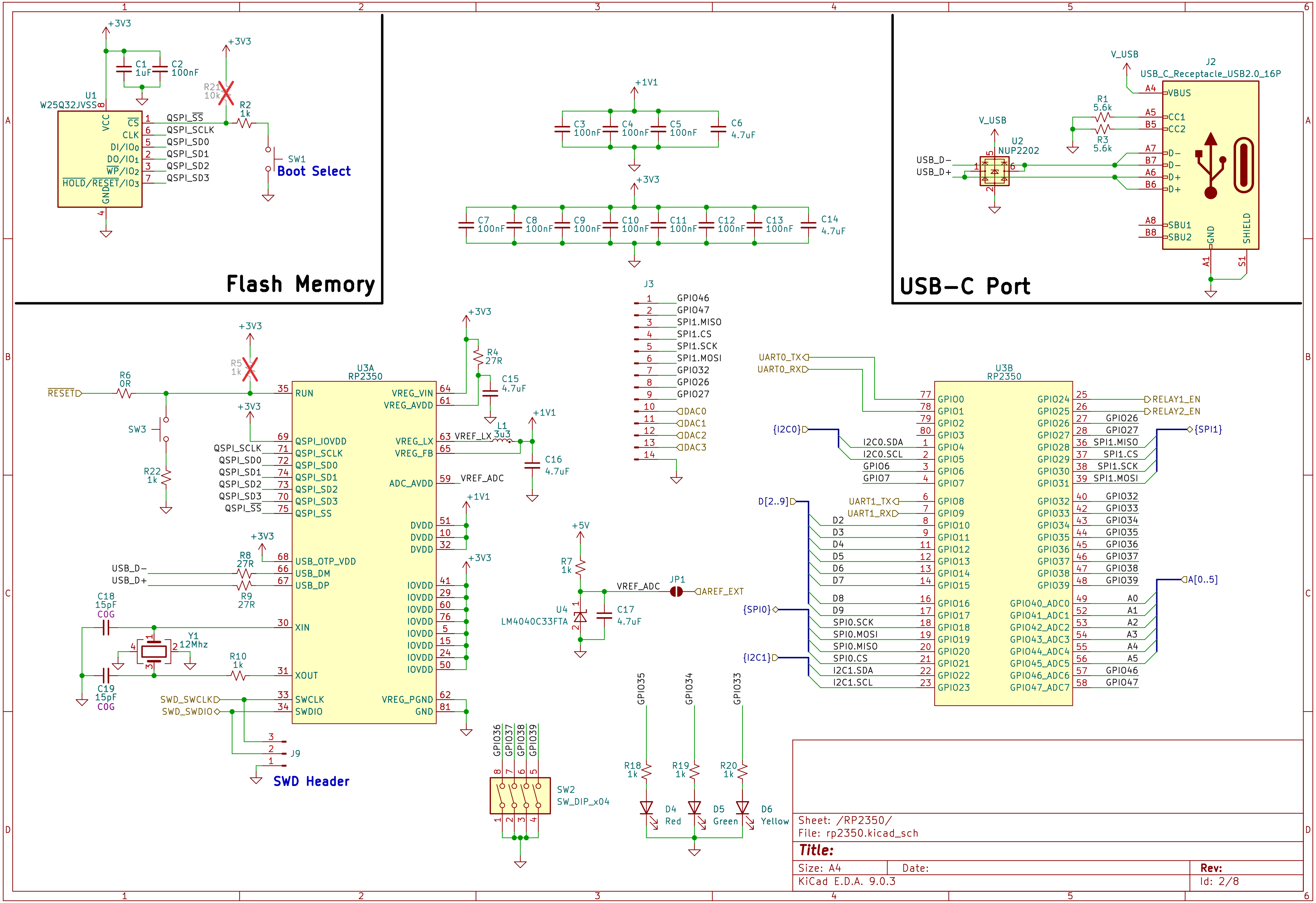
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.3		Id: 3/8


Flash Memory
U1
W25Q32JVSS
VCC
CS
CLK
DI/IO0
DO/IO1
WP/IO2
HOLD/RESET/IO3
GND
QSPI_SS
QSPI_SCLK
QSPI_SD0
QSPI_SD1
QSPI_SD2
QSPI_SD3
+3V3
C1 1uF
C2 100nF
R21 10k
R2 1k
SW1
Boot Select
+1V1
C3 100nF
C4 100nF
C5 100nF
C6 4.7uF
C7 100nF
C8 100nF
C9 100nF
C10 100nF
C11 100nF
C12 100nF
C13 100nF
C14 4.7uF
USB-C Port
V_USB
J2
USB_C_Receptacle_USB2.0_16P
VBUS
CC1
CC2
D-
D+
SBU1
SBU2
GND
SHIELD
R1 5.6k
R3 5.6k
U2
NUP2202
USB_D-
USB_D+
J3
GPIO46
GPIO47
SPI1.MISO
SPI1.CS
SPI1.SCK
SPI1.MOSI
GPIO32
GPIO26
GPIO27
DAC0
DAC1
DAC2
DAC3
U3A
RP2350
RESET
R6 0R
R5 1k
SW3
R22 1k
RUN
QSPI_IOVDD
QSPI_SCLK
QSPI_SD0
QSPI_SD1
QSPI_SD2
QSPI_SD3
QSPI_SS
USB_OTP_VDD
USB_DM
USB_DP
R8 27R
R9 27R
XIN
XOUT
C18 15pF C0G
C19 15pF C0G
Y1 12Mhz
R10 1k
SWCLK
SWDIO
SWD_SWCLK
SWD_SWDIO
J9
SWD Header
VREG_VIN
VREG_AVDD
VREG_LX
VREG_FB
ADC_AVDD
DVDD
IOVDD
VREG_PGND
GND
R4 27R
C15 4.7uF
L1 3u3
VREF_LX
C16 4.7uF
VREF_ADC
+5V
R7 1k
U4
LM4040C33FTA
C17 4.7uF
JP1
AREF_EXT
GPIO36
GPIO37
GPIO38
GPIO39
SW2
SW_DIP_x04
GPIO35
GPIO34
GPIO33
R18 1k
R19 1k
R20 1k
D4 Red
D5 Green
D6 Yellow
U3B
RP2350
UART0_TX
UART0_RX
{I2C0}
I2C0.SDA
I2C0.SCL
GPIO6
GPIO7
D[2..9]
UART1_TX
UART1_RX
D2
D3
D4
D5
D6
D7
D8
D9
{SPI0}
SPI0.SCK
SPI0.MOSI
SPI0.MISO
SPI0.CS
{I2C1}
I2C1.SDA
I2C1.SCL
GPIO0
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
GPIO8
GPIO9
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
GPIO15
GPIO16
GPIO17
GPIO18
GPIO19
GPIO20
GPIO21
GPIO22
GPIO23
GPIO24
GPIO25
GPIO26
GPIO27
GPIO28
GPIO29
GPIO30
GPIO31
GPIO32
GPIO33
GPIO34
GPIO35
GPIO36
GPIO37
GPIO38
GPIO39
GPIO40_ADC0
GPIO41_ADC1
GPIO42_ADC2
GPIO43_ADC3
GPIO44_ADC4
GPIO45_ADC5
GPIO46_ADC6
GPIO47_ADC7
RELAY1_EN
RELAY2_EN
{SPI1}
SPI1.MISO
SPI1.CS
SPI1.SCK
SPI1.MOSI
A[0..5]
A0
A1
A2
A3
A4
A5
GPIO46
GPIO47
Sheet: /RP2350/
File: rp2350.kicad_sch
Title:
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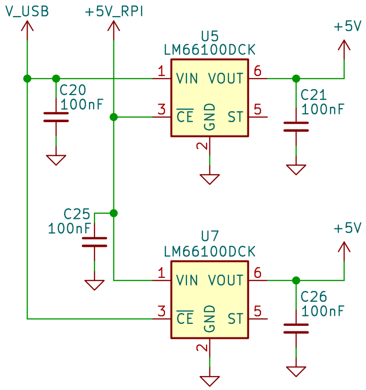
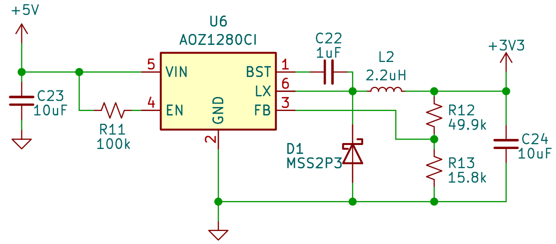
Absolute max power consumption for a PI5 is 10W. Standard pi supply is 15.3W (which we would specify as a requirement).

From here we have 5.3W, ~1A for any USB-A peripherals and our hardware.

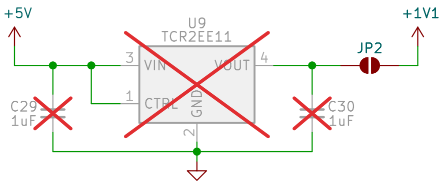
I think it's reasonable for us to assume we have around 300mA to use.



5V "or" circuit



3.3V @ 1.2A



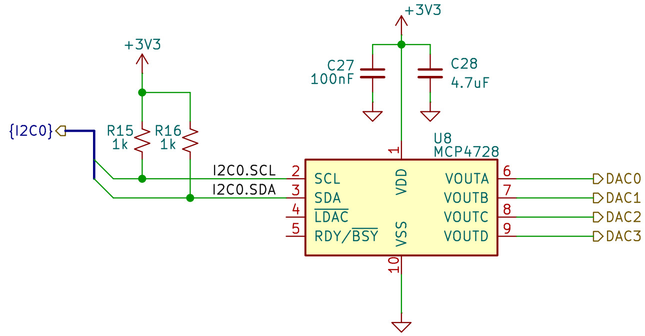
1.1V @ 200mA

Only pop as replacement for RP2350 internal Vreg

Sheet: /PMIC/
File: pmic.kicad_sch

Title:

Size: A4	Date:	Rev:
KiCad E.D.A. 9.0.3		Id: 4/8



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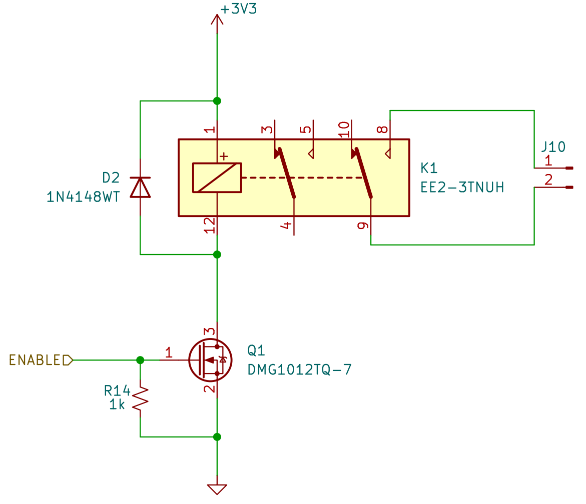
File: dac.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.3		Id: 6/8



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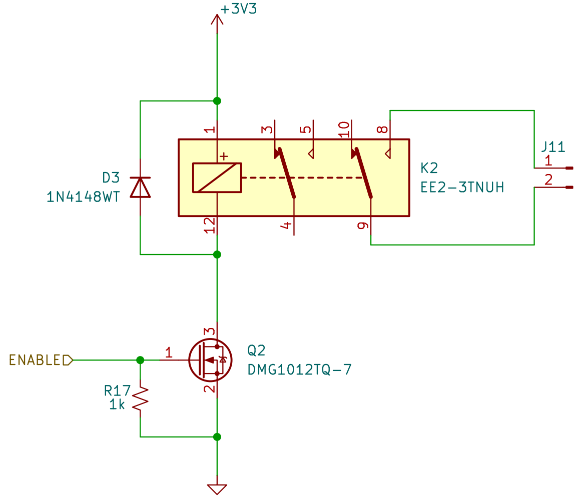
File: relay.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.3		Id: 7/8



Sheet: /Relay_2/
File: relay.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.3		Id: 8/8